

Hardware Architecture — Advanced RC Car V2

As-Built revision B (TB6612FNG rework) · single 3.3 V logic rail + 12 V motor rail · 4WD, four independent wheel channels · 38-pin ESP32 DevKit

Rev B · July 2026 · supersedes the BTS7960 As-Built revision (both original BTS7960 boards burnt out and were replaced).

What changed in Rev B (vs. the BTS7960 as-built)

- **Motor drivers replaced:** the two BTS7960 / IBT-2 H-bridges burnt out. They are replaced by **2× TB6612FNG dual H-bridge breakouts**.
- **Topology upgraded:** each TB6612 has two channels, so the car now has **one driver channel per wheel — four independently controllable motors** (previously the two motors on a side were hard-wired in parallel on one channel).
- **Motors : encoders are now 1 : 1** and index-aligned everywhere: 0=Left-Front, 1=Left-Rear, 2=Right-Front, 3=Right-Rear.
- **Speed control scheme:** each board's **PWMA/PWMB pins are tied to 3.3 V** and the IN1/IN2 pins are PWM'd instead (standard TB6612/DRV8833 configuration) — the 38-pin DevKit doesn't have enough GPIOs for dedicated PWM pins, four encoders and two STBY lines.
- **Safety lines changed:** the per-channel R_EN/L_EN enables are gone. **Each board has its own STBY pin** — LEFT board on **GPIO12**, RIGHT board on **GPIO2** — each with a **10 kΩ pull-down to GND**. STBY LOW = that board's outputs disabled (boot-safe + e-stop hardware cut).
- **Right-Front encoder moved to GPIO4/23** — this 38-pin DevKit does not expose GPIO36/39.
- **GPIO16/17 reclaimed:** the ESP32-CAM UART2 reservation was given up; those pins drive the Right-Front channel. The camera (still planned) will link over Wi-Fi instead.
- Carried over from Rev A: two power rails only (3.3 V + 12 V, no 5 V bus); AHT20 and MPU6050 intentionally excluded; encoders powered from 3.3 V, no level shifting.

⚠ **Current headroom is the #1 operating constraint now.** The TB6612FNG is rated **1.2 A continuous / 3.2 A peak per channel** (VM max 15 V — the 12 V rail is fine). A JGB37-520 stalls at roughly 3 A on 12 V — right at the peak rating. The old BTS7960s (43 A) shrugged off stall; these will not. **Ramp PWM gently, never hold a stalled wheel, keep the boards ventilated/heatsinked.** The driver has thermal shutdown, but treat that as a last resort, not a feature.

Section 1 — Components (as used)

Component	Qty	Supply	Interface / pins	Role
ESP32 DevKit (38-pin, ESP-WROOM-32)	1	3.3 V	Wi-Fi AP, PWM, I ² C, GPIO interrupts	Master MCU — runs the test-bench firmware
TB6612FNG dual H-bridge	2	12 V motor (VM) · 3.3 V logic (VCC)	Per channel: IN1/IN2 (PWM'd); PWMA/PWMB tied to 3.3 V; STBY per board	One channel per wheel — 4 independent bidirectional drives
JGB37-520 geared motor + quadrature encoder	4	12 V motor · 3.3 V encoder	2 motor leads + encoder A/B	Drive wheels with speed/position feedback

BNO055 9-axis IMU	1	3.3 V	I ² C @ 0x28	Fused absolute orientation (heading/roll/pitch)
VL53L0X ToF range sensor	1	3.3 V	I ² C @ 0x29	Distance sensing (planned subsystem)
ESP32-CAM (AI-Thinker) + OV2640	1	3.3 V	Wi-Fi link (UART pins were reclaimed)	Camera streamer (planned subsystem)
LM2596 buck converter	1	12 V in → 3.3 V out	—	Single regulated logic rail (set pot to 3.3 V)
3S Li-ion pack (3×18650)	1	11.1–12.6 V	—	Main supply
Fuse + master switch	1	12 V	—	Protection / power cutoff

AHT20 temperature/humidity sensor and MPU6050 IMU are intentionally **not** part of this build. The retired BTS7960 boards must not return to the design — the firmware no longer supports them.

Section 2 — Power Architecture (two rails)

The 3S pack feeds a fused, switched 12 V bus that splits into exactly two rails:

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Battery (11.1–12.6 V) → Fuse (15 A) → Master Switch → 12 V Bus
├── 12 V direct → TB6612 #1 VM (motor power, LEFT board: LF + LR)
├── 12 V direct → TB6612 #2 VM (motor power, RIGHT board: RF + RR)
└── LM2596 buck → 3.3 V Rail
    ├── ESP32 DevKit (3V3 pin)
    ├── ESP32-CAM (3V3 pin, planned)
    ├── TB6612 #1/#2 VCC (logic) + PWMA/PWMB (tied high)
    ├── BNO055, VL53L0X (I2C sensors)
    └── 4× encoder boards (A/B logic)

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- **12 V rail:** direct battery to both TB6612 **VM** inputs. TB6612 VM absolute max is 15 V, so a fully charged 3S pack (12.6 V) is within spec. Add $\geq 470\ \mu\text{F}$ bulk electrolytic close to each board's VM/GND.
- **3.3 V rail:** the LM2596 is adjusted to 3.3 V and powers everything logic. The ESP32 boards are fed on their 3V3 pins, bypassing their on-board regulators. TB6612 **VCC** accepts 2.7–5.5 V, so 3.3 V logic drives it natively. The PWMA/PWMB pins of both boards also connect to this rail (tied high).
- **No 5 V bus.** Nothing in this build requires 5 V.

Buck sizing: set the LM2596 to 3.3 V before connecting any load and verify under load. The 3.3 V rail draws $\approx 0.7\ \text{A}$ peak (Section 7); the LM2596's 3 A rating leaves ample margin. Add $\geq 470\ \mu\text{F}$ input and $\geq 220\ \mu\text{F}$ output bulk capacitors plus $0.1\ \mu\text{F}$ ceramics.

Feeding ESP32 via 3V3: powering the 3V3 pin bypasses the on-board regulator and reverse-protection, so the 3.3 V rail must be clean and never exceed 3.6 V. Do not also power via USB and VIN at the same time.

Grounding & noise

- Single star-ground near the battery negative. 12 V motor returns 14–16 AWG; logic/sensor grounds 22–24 AWG back to the same node.

- 0.1 μF ceramics at each IC power pin; 100–470 μF electrolytic on the 12 V bus near the drivers and on the 3.3 V rail near the ESP32.
- Twist motor + encoder wires; keep I²C (4.7 k Ω pull-ups to 3.3 V) away from motor power.

Section 3 — GPIO Planning (38-pin ESP32 DevKit master)

GPIO	Function	Dir	Notes
18	TB6612 #1 AIN1 — Left-Front dir/PWM 1	PWM out	20 kHz
19	TB6612 #1 AIN2 — Left-Front dir/PWM 2	PWM out	20 kHz
26	TB6612 #1 BIN1 — Left-Rear dir/PWM 1	PWM out	20 kHz
27	TB6612 #1 BIN2 — Left-Rear dir/PWM 2	PWM out	20 kHz
16	TB6612 #2 AIN1 — Right-Front dir/PWM 1	PWM out	reclaimed from CAM UART
17	TB6612 #2 AIN2 — Right-Front dir/PWM 2	PWM out	reclaimed from CAM UART
13	TB6612 #2 BIN1 — Right-Rear dir/PWM 1	PWM out	20 kHz
15	TB6612 #2 BIN2 — Right-Rear dir/PWM 2	PWM out	strapping pin (MTDO) — boot glitch masked by STBY pull-down
12	STBY — TB6612 #1 (LEFT board)	out	10 kΩ pull-down to GND required. MTDI strapping pin must be LOW at boot anyway — the pull-down doubles as the motors-dead-at-boot guarantee
2	STBY — TB6612 #2 (RIGHT board)	out	10 kΩ pull-down to GND required. Boot strap wants low (helps flashing); drives the onboard LED — lit = right board enabled
34 / 35	Encoder 1 A/B — Left-Front	in (IRQ)	input-only, external pull-up to 3.3 V
32 / 33	Encoder 2 A/B — Left-Rear	in (IRQ)	external pull-up to 3.3 V
4 / 23	Encoder 3 A/B — Right-Front	in (IRQ)	moved from 36/39 (not present on this 38-pin board); external pull-up to 3.3 V
25 / 14	Encoder 4 A/B — Right-Rear	in (IRQ)	external pull-up to 3.3 V
21 / 22	I ² C SDA / SCL	I/O	BNO055 (0x28), VL53L0X (0x29); 4.7 kΩ pull-ups to 3.3 V
0	Boot strapping / BOOT button	—	left unused for I/O
5	spare	—	strapping pin; left unused
1 / 3	UART0 (USB serial)	—	flashing + serial monitor
6–11	SPI flash	—	do not use
36 / 39	—	—	not exposed on this board — unused

3.3 V everywhere means encoder A/B pulses are already at logic level and connect straight to the ESP32 inputs with no dividers or level shifters.

Boot safety: GPIO15 (an IN pin) can glitch during the ~300 ms of boot before firmware runs. This is harmless because each pulled-down STBY holds its bridge off until the firmware takes control. Those two 10 kΩ resistors (GPIO12 and GPIO2 → GND) are the entire boot-spin fix — do not omit them.

Section 4 — Drive Topology (4WD, four independent channels)

Four motors, two driver boards, **four control channels**. Each TB6612FNG contains two full H-bridges (channel A and channel B), so each wheel gets its own bidirectional drive. With PWMA/PWMB tied to 3.3 V, direction *and* speed both come from the IN pins: duty on IN1 = forward, duty on IN2 = reverse. Steering is differential (skid-steer), and per-wheel control additionally allows torque vectoring, per-wheel testing and slip diagnosis.

Index / key	Wheel	Driver channel	GPIO (IN1/IN2)	STBY	Encoder (feedback)
0 · lf	Left-Front	TB6612 #1, ch A	18 / 19	12	Enc 1 (34/35)
1 · lr	Left-Rear	TB6612 #1, ch B	26 / 27	12	Enc 2 (32/33)
2 · rf	Right-Front	TB6612 #2, ch A	16 / 17	2	Enc 3 (4/23)
3 · rr	Right-Rear	TB6612 #2, ch B	13 / 15	2	Enc 4 (25/14)

Motors and encoders are **1 : 1 and index-aligned** — motor *i* is measured by encoder *i* in firmware, REST API and telemetry alike.

Channel truth table (PWM pin tied high, board STBY high)

IN1	IN2	Result
PWM duty	L	Forward (off-phase = coast / fast decay)
L	PWM duty	Reverse
H	H	Short brake
L	L	Coast — outputs off ("stop")
STBY = LOW		Whole board off (master cut)

Section 5 — Wiring (per component)

TB6612FNG #1 — LEFT board (LF + LR)

- AIN1→GPIO18, AIN2→GPIO19 (Left-Front) · BIN1→GPIO26, BIN2→GPIO27 (Left-Rear)
- **PWMA and PWMB → 3.3 V rail (tied high)**
- STBY → GPIO12, with 10 kΩ pull-down to GND
- VCC (logic) → 3.3 V rail · GND → common ground · VM → 12 V bus ($\geq 470\ \mu\text{F}$ bulk cap at the board)
- A01/A02 → Left-Front motor · B01/B02 → Left-Rear motor

TB6612FNG #2 — RIGHT board (RF + RR)

- AIN1→GPIO16, AIN2→GPIO17 (Right-Front) · BIN1→GPIO13, BIN2→GPIO15 (Right-Rear)
- **PWMA and PWMB → 3.3 V rail (tied high)**
- STBY → GPIO2, with 10 kΩ pull-down to GND (onboard LED shows board-enabled state)
- VCC → 3.3 V · GND → common · VM → 12 V (+bulk cap)
- A01/A02 → Right-Front motor · B01/B02 → Right-Rear motor

If a wheel spins the wrong way, swap that motor's two leads (O1/O2) at the driver — do not change the firmware convention.

Motors & encoders (×4)

- Each motor's leads → its own TB6612 output pair (12 V pulsed) — **no motors in parallel any more**
- Encoder Vcc → 3.3 V rail · Encoder GND → ground · A/B → assigned GPIO (Section 3). External pull-ups to 3.3 V on all encoder lines (34/35 are input-only and have no internal pull-ups; use the same harness pull-ups everywhere for consistency). Never feed 12 V to an encoder.

Sensors (I²C)

- BNO055: Vcc→3.3 V, GND, SDA→21, SCL→22, ADR/SDO→GND (0x28)
- VL53L0X: Vcc→3.3 V, GND, SDA→21, SCL→22 (0x29), XSHUT→3.3 V to enable

ESP32-CAM (planned)

- 3V3 → 3.3 V rail · GND → ground · **link over Wi-Fi** (the ESP32-CAM has its own radio). The old wired UART2 link (GPIO16/17) is gone — those pins now drive the Right-Front channel.

Section 6 — Signal Flow

ESP32 DevKit —(IN1/IN2 PWM ×4, STBY ×2)→ TB6612FNG ×2 —(12 V PWM)→ 4 motors (one channel each)
Wheel encoders —(A/B quadrature, IRQ)→ ESP32 DevKit
BNO055, VL53L0X —(I²C @ 3.3 V)→ ESP32 DevKit
ESP32-CAM (OV2640) —(Wi-Fi)→ phone / DevKit [planned]
ESP32 DevKit —(Wi-Fi AP, HTTP + WebSocket)→ phone test UI

Section 7 — Power Budget

Load	Rail	Typ	Peak	Notes
ESP32 DevKit	3.3 V	80 mA	240 mA	Wi-Fi AP active

ESP32-CAM	3.3 V	180 mA	310 mA	camera + Wi-Fi (planned)
BNO055	3.3 V	12 mA	15 mA	fusion mode
VL53L0X	3.3 V	19 mA	40 mA	active ranging
4× encoders	3.3 V	20 mA	40 mA	hall sensors
2× TB6612 logic (VCC + PWM ties)	3.3 V	≈4 mA	≈10 mA	far lighter than the BTS7960 gate drive
3.3 V rail total	3.3 V	≈0.32 A	≈0.66 A	well within LM2596 3 A
4× DC motors	12 V	~2–6 A	~12.8 A (all stalled)	stall ~3.2 A each — at the TB6612 per-channel peak limit

Motor current vs. driver rating: each channel must stay ≤ 1.2 A continuous. A JGB37-520 stall (~ 3.2 A) touches the TB6612's absolute peak — survivable for milliseconds, destructive if held. The 15 A pack fuse protects the wiring, **not** the drivers. Operate with soft PWM ramps and cut drive (e-stop / coast) the moment a wheel stops turning under power. Consider a small heatsink + airflow on both boards.

- 12 V rail: keep the 15 A fuse and a pack rated ≥ 20 –30 A burst.
- 3.3 V rail: LM2596 input current at 12 V for 3.3 V @ 0.7 A out ≈ 0.25 A; the buck dissipates only ~ 1 W.
- Battery: a 5–6 Ah 3S pack gives ~ 1.5 –2 h moderate runtime.

Section 8 — Firmware Mapping

This hardware is exercised by the **CarTestBench** ESP32 firmware. Each subsystem has its own page on the phone UI (the ESP32 hosts a Wi-Fi access point — connect and browse to its IP):

Page	Hardware tested	Status
/motors	All four TB6612 channels independently (per-wheel arm, PWM, brake/coast) + ARM ALL / master e-stop	implemented
/encoders	4 wheel encoders + CPR calibration	implemented
/imu	BNO055 orientation	groundwork (feature-flagged)
Range	VL53L0X	planned
Camera	ESP32-CAM / OV2640 (Wi-Fi link)	planned

- REST API: `/api/motor?ch=lf|lr|rf|rr|all` with `arm=`, `pwm=-255..255`, `mode=brake|coast|disarm`; master `/api/estop`.
- Telemetry (WebSocket, 10 Hz): `m:[pwm×4]`, `a:[armed×4]`, `enc:[{c,r}×4]` — index order LF, LR, RF, RR.
- Safety model: each board's STBY is LOW at boot and whenever neither of its channels is armed; a master e-stop drops both boards (hardware cut). A disarmed channel additionally holds `IN1=IN2=LOW`.

Pin assignments above are the single source of truth and match `config.h` in the firmware (`PINS_MOTOR[]`, `PIN_MOTOR_STBY_LEFT/RIGHT`, `PINS_ENCODER[]`).